CCCA Meta-Atlas — Caron et al. 2020 (Acute Lymphoblastic Leukemia)

Pediatric B/T-cell ALL · 10x · 25,788 malignant blasts · 8 samples, 3 genetic subtypes (ETV6-RUNX1, HHD, pre-T)

GO / NO-GO GATES

Gate 1 (mixing + continuity) — GO

Harmony(theta=10) lifts per-neighbourhood iLISI 1.0 -> 4.7 of 8 samples; the 3 genetic subtypes intermix on one connected manifold (single component after island removal). Read on the embedding + iLISI sweep, not a scalar.

Gate 2 (module geography) — GO (non-branching)

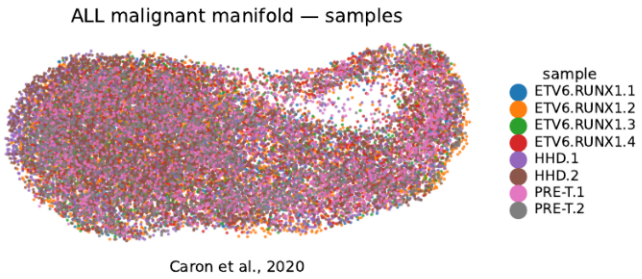
PPT tree = 0 forks -> non-branching. Organized Gast<->pSom cross-repression (r=-0.45; -0.37 after removing cell cycle) + Crest inverse gradient (r=-0.72 vs pSom) SURVIVE cell-cycle regression -> real developmental axis, not a proliferation artefact.
FLAG: Crest leans to the Gast end (inversion of ideal pSom-arm), passes anyway.

Gate 3 (cross-repression) — NA

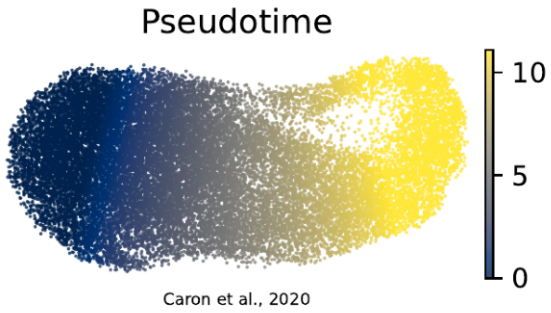
Non-branching topology has no bifurcation to test (branching-only gate).

STEP FIGURES

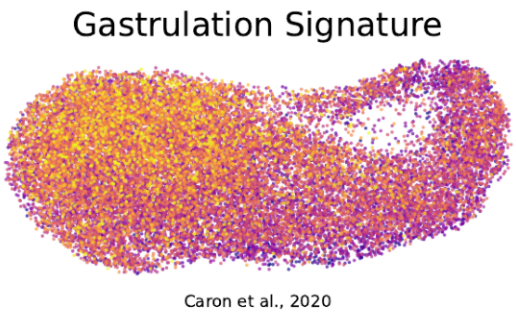
Samples (Gate 1: integration)



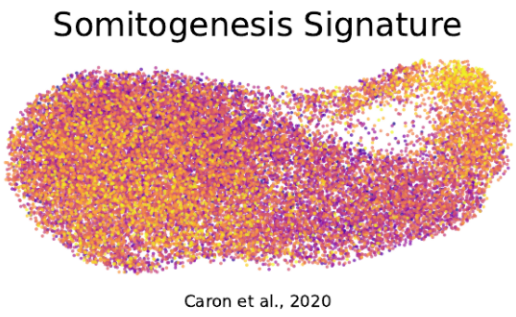
Manifold (pseudotime)



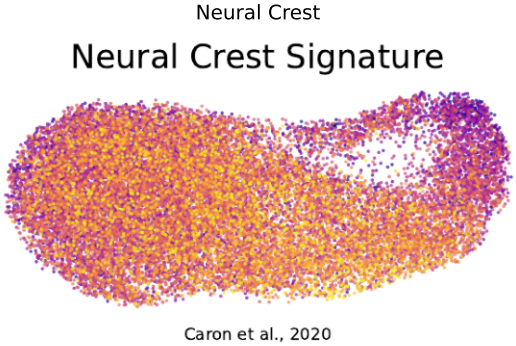
Gastrulation



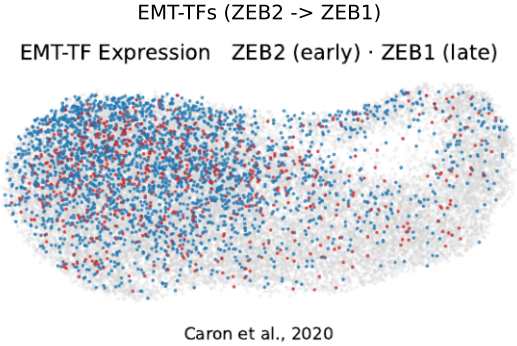
Somitogenesis



Neural Crest



EMT-TFs (ZEB2 -> ZEB1)



KEY PARAMETERS & RATIONALE

Threshold: none — 8 samples span 1,585–5,816 cells with no clean multiplicative cliff (>3×); gap rule is a no-op (keep all).
Embedding: n_pcs=20, UMAP spread=1.3 (10x precedent); Harmony theta=10 (iLISI plateaus by theta=5; subtypes need strong integration).
Geometry: NON-BRANCHING principal curve (ElPiGraph, fold-free self-approach 3.0); PPT tree gave 0 branch nodes.
Root: quiescent Gast/pSom-high pole (developmental origin; Crest is late). Hub / Gate-3 r: NA (non-branching).
EMT-TFs (data-derived, wide panel incl. FOXC/PRRX): ZEB2 (root) → ZEB1 (terminus); other factors <3% detection in blasts.
Caveat: dominant axis is the CELL CYCLE (pseudotime ρ=+0.69 with cycling; cc explains R^2=0.37/0.17/0.11 of Gast/pSom/Crest); the developmental cross-repression is the residual structure surviving cell-cycle removal (Zhou 2020 precedent).

VERDICT: Pattern reproduces as a non-branching developmental axis — organized Gast↔pSom cross-repression with a terminal Crest gradient, cell-cycle-independent. G1 GO / G2 GO / G3 NA. Crest-arm inversion flagged.